COURSE-BASED PROJECT DOCUMENTATION | DOC-09

Java 21 Language Features Supported

Modern Java 21 LTS Syntax Conformance & Grammar Rules

1. Supported Java 21 Features (Verified in Repository)

The parser implements full grammatical support for modern Java 21 LTS finalized syntax:

Feature Category	Java 21 Syntax Example	Test Verification File
Record Declarations	public record Point(int x, int y) {}	tests/valid_full.java
Compact Record Const.	public Point { Objects.requireNonNull(x...	tests/test_record_compact...
Record Patterns (JEP 440)	p instanceof Point(var x, var y)	tests/java21_record_patte...
Nested Record Patterns	p instanceof Box(Point(int x, int y))	tests/java21_record_patte...
Switch Patterns (JEP 441)	case Point(double x, double y) -> ...	tests/java21_switch_patte...
Switch Guards (when)	case Point(int x, int y) when x > 0 ->	tests/java21_switch_patte...
Sealed Types	sealed interface Shape permits Circle {}	tests/test_advanced.java
Text Blocks	String s = """ Hello World """;	tests/err_text_block.java
Unsigned Right Shift	a >>>= 2;	tests/test_urshift_assign...
Module Info	module com.app { requires java.base; }	tests/test_module_info.java
Ragged Arrays	int[][] matrix = new int[5][];	tests/test_array_ragged.java
Multi Try-with-Resources	try (R r1 = a; R r2 = b;) { ... }	tests/test_try_resources_...

2. Code Snippet: Java 21 Pattern Matching with Guard

```
public class PatternDemo {
    public record Point(double x, double y) {}

    public static String classify(Object obj) {
        return switch (obj) {
            case null -> "Null reference";
            case Point(double x, double y) when x > 0 && y > 0 -> "Quadrant I Point";
            case Point(double x, double y) -> "Other Point";
            case String s -> "String: " + s;
            default -> "Unknown Object";
        };
    }
}
```